WORKED SOLUTION

STEP I 2011, Question 4

Pure • Tangents to a Parabola

STEP I 2011, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s1-q4

The key idea

Write the tangent at a general point of the parabola in a form with no fractions, and the intersection of two tangents drops out by subtraction. For the angle, use the scalar product of $\overrightarrow{FT}$ and $\overrightarrow{FP}$. The lengths are kinder than they look: FP is the focal distance, and FT^2 factorises. Once $\cos\phi$ is found, notice that the answer is symmetric in p and q. That symmetry is the whole of the bisection argument.

Solution

The point T

Differentiating $y^2 = 4ax$ implicitly gives $2y\dfrac{\mathrm{d}y}{\mathrm{d}x} = 4a$, so at P, where $y = 2ap \neq 0$ (because $p \neq 0$), the gradient is $\dfrac{4a}{4ap} = \dfrac{1}{p}$. The tangent at P is therefore

$$y - 2ap = \frac{x - ap^2}{p} \quad \Longleftrightarrow \quad py = x + ap^2,$$

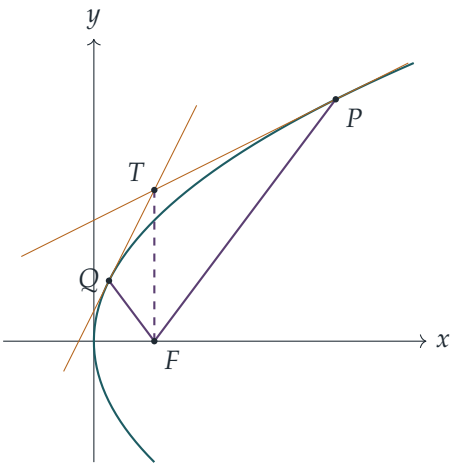
and likewise the tangent at Q is $qy = x + aq^2$. Subtracting,

$$(p - q)\,y = a(p^2 - q^2) = a(p - q)(p + q).$$

Since P and Q are distinct, $p \neq q$, so we may divide by $p - q$ to get $y = a(p + q)$. Then

$$x = py - ap^2 = ap^2 + apq - ap^2 = apq,$$

so T is the point $(apq,\, a(p + q))$, as required.



The figure has $a = 1$, $p = 2$, $q = \frac{1}{2}$, where FT happens to be vertical.

The angle ϕ

Note first that $T \neq F$: that would need $p + q = 0$ and $pq = 1$, so $p^2 = -1$, which is impossible. So the angle TFP makes sense. With $F = (a, 0)$,

$$\overrightarrow{FP} = a\begin{pmatrix} p^2 - 1 \\ 2p \end{pmatrix}, \qquad \overrightarrow{FT} = a\begin{pmatrix} pq - 1 \\ p + q \end{pmatrix}.$$

The lengths are

$$FP^2 = a^2[(p^2-1)^2 + 4p^2] = a^2(p^2+1)^2,$$
$$FT^2 = a^2[(pq-1)^2 + (p+q)^2] = a^2[p^2q^2 + p^2 + q^2 + 1] = a^2(p^2+1)(q^2+1),$$

since the cross terms $-2pq$ and $+2pq$ cancel. So $FP = |a|(p^2+1)$ and $FT = |a|\sqrt{(p^2+1)(q^2+1)}$.

Ask yourself How do we spot that the scalar product will factorise?

We expect (p^2+1) to cancel with FP, so we look for it. Expanding,

$$\begin{aligned} \overrightarrow{FP} \cdot \overrightarrow{FT} &= a^2[(p^2-1)(pq-1) + 2p(p+q)] \\ &= a^2[p^3q - p^2 - pq + 1 + 2p^2 + 2pq] \\ &= a^2[p^3q + pq + p^2 + 1] \\ &= a^2(p^2+1)(pq+1). \end{aligned}$$

Therefore, using $|a|^2 = a^2$,

$$\cos\phi = \frac{\overrightarrow{FP} \cdot \overrightarrow{FT}}{FP \cdot FT} = \frac{a^2(p^2+1)(pq+1)}{a^2(p^2+1)\sqrt{(p^2+1)(q^2+1)}} = \frac{pq+1}{\sqrt{(p^2+1)(q^2+1)}},$$

which is the required result.

The bisection

The coordinates of T are symmetric in p and q, so swapping the roles of P and Q in the calculation above shows that the angle TFQ has

$$\cos\angle TFQ = \frac{qp+1}{\sqrt{(q^2+1)(p^2+1)}} = \cos\phi.$$

Both angles lie in $[0,\pi]$, where cosine is one-to-one, so $\angle TFP = \angle TFQ = \phi$.

So the rays FP and FQ each make the angle ϕ with the ray FT. In the plane there are only two rays from F making angle ϕ with FT, and they are mirror images in the line FT. The rays FP and FQ are not the same ray: a line through the focus meets the parabola in two points on opposite sides of F (or only once), so one ray from F cannot pass through both P and Q. Hence FP and FQ are reflections of each other in the line FT, which is exactly the statement that the line FT bisects the angle PFQ.

One detail explains why the question says the *line* FT. If $pq+1>0$ then $\phi<\frac{\pi}{2}$, and the ray FT itself lies inside the angle $PFQ = 2\phi$. If $pq+1<0$ then $\phi>\frac{\pi}{2}$, the angle PFQ is $2\pi-2\phi$, and it is the extension of TF beyond F that bisects it. If $pq=-1$ then PQ is a focal chord, $\phi=\frac{\pi}{2}$, and FT is perpendicular to it, bisecting the straight angle. In every case the line FT bisects the angle PFQ.

Checks

Take $a=1$, $p=2$, $q=\frac{1}{2}$ as in the figure: $T=(1,\frac{5}{2})$, $\overrightarrow{FP}=(3,4)$, $\overrightarrow{FQ}=(-\frac{3}{4},1)$ and $\overrightarrow{FT}=(0,\frac{5}{2})$. Then $\cos\angle TFP=\frac{4}{5}$ and $\cos\angle TFQ=\frac{1}{5/4}=\frac{4}{5}$, matching $\dfrac{2}{\sqrt{5\cdot\frac{5}{4}}}=\dfrac{4}{5}$.

A computer testing 10^5 random choices of a (of either sign), p and q confirmed the coordinates of T, the formula for $\cos\phi$ and the bisection to within 10^{-10}. It included many cases with $pq<-1$, where the backward extension of FT does the bisecting.

Common slips

- Dividing by $p-q$ without saying why it is non-zero (the points are distinct).
- Writing $FP=a(p^2+1)$, which is wrong if $a<0$. Only a^2 appears in $\cos\phi$, so the sign of a does not matter in the end.
- Saying "the cosines are equal, so FT bisects" without explaining why FP and FQ lie on opposite sides of the line FT.

Beyond STEP

This is the reflective property of the parabola in disguise: the focus sees the two tangent segments from any external point at equal angles. The same holds for the ellipse and hyperbola, and all three come from the focus–directrix definition of a conic, which

first-year geometry and projective geometry courses develop much further. For more questions in this spirit, Siklos's *Advanced Problems in Mathematics* is free and well worth working through.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.